Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2024

Course: B.Tech		*Time: 03 Hours*
Code: 100310	*Subject: Engineering Mechanics*	*Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Which of the following is a second-order tensor?
- (i) Scalar
- (ii) Stress tensor
- (iii) Vector
- (iv) Displacement

(b) The principal axes of a tensor correspond to:
- (i) Directions of zero stress
- (ii) Eigenvectors of the tensor
- (iii) Directions where the tensor is anti-symmetric
- (iv) Directions perpendicular to force

(c) In a simply supported beam with a central point load, the maximum bending moment occurs
- (i) At the supports
- (ii) At the point of load application
- (iii) At the mid-span
- (iv) At quarter-span

(d) In a cantilever beam with a point load at the free end, the maximum bending moment occurs at
- (i) Free end
- (ii) Midpoint
- (iii) Fixed end
- (iv) Quarter-span

(e) In a shear force diagram (SFD), a sudden jump indicates
- (i) A uniformly distributed load
- (ii) A uniformly varying load
- (iii) A point load
- (iv) No loading

(f) The unit of the modulus of rigidity G in SI system is
- (i) Pascal (Pa)
- (ii) Meter (m)
- (iii) Newton (N)
- (iv) Joule (J)

(g) In a solid circular shaft under torsion, where does the maximum shear stress occur?
- (i) At the center
- (ii) At the axis
- (iii) At the outer surface
- (iv) Uniformly throughout

(h) Which property is considered when designing a shaft based on strength?
- (i) Torsional rigidity
- (ii) Lateral rigidity
- (iii) Yield strength
- (iv) Flexibility

(i) The force of friction always acts
- (i) In the direction of motion
- (ii) Perpendicular to the surface
- (iii) Opposite to the direction of motion or tendency of motion
- (iv) Along the normal to the surface

(j) Angle of repose is the angle at which
- (i) The body starts rotating
- (ii) A body just begins to slide down an inclined plane
- (iii) The friction becomes zero
- (iv) Normal force equals weight

Q.2 (a) What is a tensor? How does it differ from a vector? Explain using indicial notation and describe the transformation law for a second-order tensor. [7]

(b) How do vectors and tensors transform under a rotation of the coordinate system? [7]

Q.3 (a) What is a rigid body? How can its motion be described using a moving coordinate system attached to it? [7]

(b) State the five-term acceleration formula for motion relative to a rotating rigid body. Explain each term briefly. [7]

Q.4 (a) State and prove the parallel axes theorem in the determination of moment of inertia of areas with the help of a neat sketch. [7]

(b) Calculate the mass moment of inertia for solid cylinder about an axis passing through its centre and perpendicular to its own axis of cylindrical symmetry. [7]

Q.5 (a) Describe the phenomenon of free precession in a gyroscope. [7]

(b) Define beam. Sketch different types of beams, types of supports and types of loads indicating their names. [7]

Q.6 (a) A 6 m long cantilever beam carries loads of 5 kN, 8 kN and 15 kN at 1m, 2.5 m and 5 m respectively from the free end and a uniformly distributed load of 12kN/m over a length of 4 m from the fixed end. Draw shear force and bending moment diagrams. [7]

(b) Derive the torsion equation $\frac{T}{J} = \frac{\tau}{r} = \frac{G\theta}{L}$ and state the assumptions involved. [7]

Q.7 (a) Derive the formula for angle of twist in a hollow circular shaft under torsion and compare it with that of a solid shaft. [7]

(b) A shaft is required to transmit 50kW power at 400 r.p.m. It is hollow shaft having inside diameter 0.6 times the outside diameter. It is made of plain carbon steel and the permissible shear stress is 84 N/mm2. Find the inner and outer diameter of the shaft nearly. [7]

Q.8 (a) State and explain Coulomb's laws of friction. How do they apply to the motion of a body on an inclined plane? [7]

(b) A body resting on a rough horizontal plane required a pull of 18 N inclined at 30° to the plane just to move it. It was found that a push of 22 N inclined at 30° to the plane just moved the body. Determine the weight of the body and coefficient of friction. [7]

Q.9 (a) A simply supported beam of length l is loaded by a uniformly distributed load w over the entire spam, it is propped at the mid span so that the deflection at the centre is zero, calculate the reaction at the prop. [7]

(b) Define the following terms [7]

(i) Torsion

(ii) Torsional rigidity

(iii) Moment of inertia

(iv) Polar Moment of inertia